Reward Shaping Is Noise Shaping: where the RLVR project stands

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1. The question

RL with verifiable rewards (RLVR) trains a language model on math: it writes answers, a checker grades them, and the update $g = \sum_i A_i \nabla \log \pi(y_i)$ pushes each answer up or down by its *advantage* A_i. Dozens of papers change the advantage (reward right answers only, PSR; penalize wrong ones only, NSR; change the baseline; aim at pass@k; add entropy or "unlikeliness" bonuses) and report more *diversity*, meaning higher pass@k (at least one of k samples right). Almost all of them compare methods **at the same training step**. Our question: is that diversity real, or is it an artifact of how methods are compared?

2. The idea and the diagnosis (finished, solid)

Theory. For any rule that scores answers only by correctness, with p a question's success rate, $\mathbb{E}g = c(p)\nabla p$ and $\mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p)$. Every such rule pushes the model in the *same direction* (∇p); it can only reweight questions and change the *noise*. Wrong answers carry most of the noise (2–3× larger squared gradients, measured on two model families). **Consequence.** Noise sets how *fast* a run moves along one accuracy–diversity curve, not the curve. A slower run is less sharpened at any fixed step, so it *looks* more diverse. The fair comparison is **pass@k at matched pass@1**, interpolated along the reference run's trajectory. **Identity.** At matched pass@1, pass@$k = \mathbb{E}_x[1-(1-p_x)^k]$, so diversity at equal accuracy only measures how evenly success is spread across questions. To move it, a method must bring signal to *rarely solved* questions.

Evidence (Qwen2.5-Math-1.5B and OLMo-2-1B, all pre-registered, reviewed by an independent model, Codex):

1. **One curve.** PSR, NSR, five baselines between them, the NSR paper's own implementation, and a pass@4 objective all move along one curve. On OLMo, NSR minus PSR at matched accuracy is -0.30, 90% $[-1.28, +0.77]$: equivalent within ± 2.
2. **Noise is the cause.** Adding zero-mean random noise of NSR's size to PSR makes it exactly as slow as NSR (speed $\phi = -0.11$ $[-0.22, -0.05]$ on a scale where NSR = 0, PSR = 1). With Adam, noise does not shrink steps; it misdirects them.
3. **Not temperature.** RL barely changes entropy on fixed text but cuts it 3–4× on its own samples; sampling a late checkpoint hotter does not beat the training curve.
4. **Popular fixes stay on the curve:** entropy bonus, adaptive entropy, entropy floors (these even stop learning), the published unlikeliness reward.

3. Turning the diagnosis into a method (so far: small gains)

The diagnosis says the only lever is *new information on rarely solved questions.* Results, as pass@16 gain over Dr. GRPO at matched pass@1 (3 seeds, 511 held-out problems):

Method	Gain	Status
Reward shaping, entropy control, unlikeliness, pass@k objectives	≈ 0	finished
Replay the model's own past successes, weighted to rare questions (Pre-reg. 12)	$+1.19$ $[+0.54, +1.79]$	finished; below the +2 bar
Same replay, seeded with base-model successes (Option A, Pre-reg. 14)	about +1; 5 points less accurate at step 150	interim
Fine-tune on the base-model successes first, then Dr. GRPO (a control in Option A)	beyond Dr. GRPO's range	interim

The surprise is the warm start: step-150 accuracy 51–53% vs 44%, pass@16 78–80 vs 75, greedy test accuracy 54.9% vs 47.5%. Its pass@16 gain sits on the hardest questions, those the base model *never* solves in 32 tries (24 → 34), exactly where the diagnosis says diversity must come from. Two caveats: (i) Dr. GRPO never gets above about 45% in 150 steps, so the warm start cannot yet be compared at matched accuracy; (ii) our 512-token answer limit truncates 48% of Dr. GRPO's answers but 33% of the warm start's, so part of the gain may be "learned to finish in time".

4. The 2026 literature (searched October 21)

- **Still ours, unpublished:** noise sets the speed along one curve; NSR on PSR's curve on two families; the noise-injection causal test; the matched-accuracy protocol; temperature does not beat the curve; the spread identity.

No 2026 method paper compares at matched accuracy, and several (BBG, EliteReplay) gain pass@k while losing pass@1, the confound we describe.

- **Already published:** "same direction, per-question weight" (Davis & Recht 2025; Heckel et al. 2026); base-seeded replay (TAR, Oct. 2026); nearly our coverage weight (BBG, June 2026); fine-tuning on own successes before RL (STaR, DART-Math, TREK, PKR); rescuing all-wrong groups (LSPO). Barakat et al. (NeurIPS 2026) show weights can rotate the update through shared parameters, so "one curve" must be an empirical claim, not a theorem.

5. Honest assessment

Independent review (Codex): **5/10** today (diagnosis strong; method gains small and partly published). A clean method win with all controls: about 7/10. Codex's brainstorm of three new RL methods (inserting base successes into all-wrong groups; noise-minimizing weights; value-of-information rollout allocation) found each already published or likely to score only 4–5/10 even if it wins. **The open problem is the method**: the diagnosis is sound, but the obvious ways of acting on it are crowded.

6. What is running and what comes next

- **Option A evaluation** (AWS): about 8 of 23 lanes left; the registered verdict follows, then Codex review. A full win is unlikely given the accuracy gap.
- **Audit of 2026 methods at matched accuracy** (AWS): TAR, BBG, MaxRL (ICML 2026), plus W-REINFORCE as BBG's control, 3 seeds each; waiting for spot GPUs. If their gains vanish at matched accuracy, the diagnosis is shown on this year's papers; if they hold, they are the baselines to beat.
- **Token-budget check** (AWS): Dr. GRPO and the warm start re-evaluated at 1,024 tokens; reading rules fixed in advance.
- **Tamia** (your H100 nodes, set up today): first, retrain Dr. GRPO and the warm start at 2,048 tokens to settle the budget question; then scale to Qwen2.5-Math-7B (BBG's own setting) to replicate the diagnosis and the audit. Scale was Codex's other main criticism.

The decision ahead: find a method that acts on "information on rare questions" inside RL and is not in the 2026 papers, or build the paper around the diagnosis plus the matched-accuracy audit of 2026 methods at two scales. If the warm-start effect survives the 2,048-token and 7B tests, it tells us exactly what an RL method has to reproduce.